Compact 2-manifold recognition

data: K_0 = Vertices
K_1 = edges (pairs of verts)
K_2 = two-cells (defined by cyclic ordering of edges)

Q: which 2-mfd do I have?

① Compute Euler char

$$\chi = |K_0| - |K_1| + |K_2|$$

② Determine Orientability

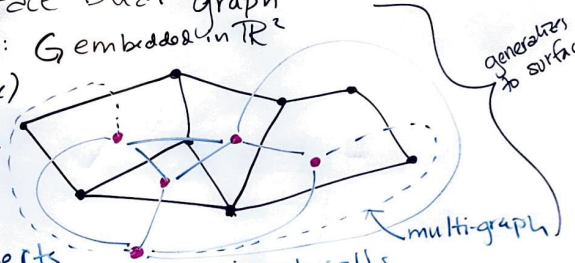
⓪ Check every edge sees exactly 2 two-cells. } is it a mfd?
↳ suff. for simp. cplx.
for general CW complex, if large enough, check links.

① Compute "Surface Dual" graph

Def: Planar Dual: G embedded in $\mathbb{R}^2$ (no deg 1 vertex)

a) put a vertex in each 2-cell

b) connect verts if they correspond to adjacent cells



G	G^*
vertex	2-cell
edge	edge
2-cells	vertex

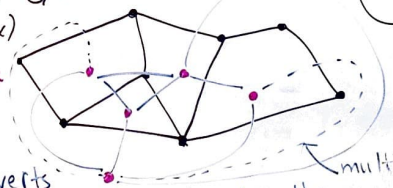
② Pick a dual vertex (2-cell in primary) and give it an orientation

③ Do BFS (or local) to propagate orientation } can be done together

④ Check every edge

① Compute "Surface Dual" graph

Def: Planar Dual: G embedded in $\mathbb{R}^2$ (no deg 1 vertex)



generators to surfaces

multigraph

a) put a vertex in each 2-cell

b) connect verts if they correspond to adjacent cells

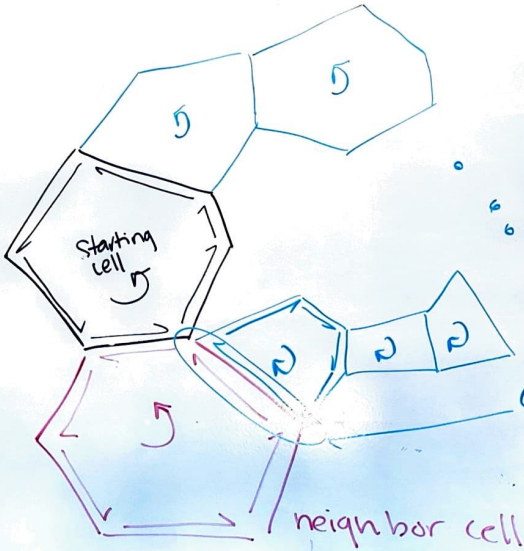
G	G^*
vertex	2-cell
edge	edge
2-cells	vertex

② Pick a dual vertex (2-cell in primary) and give it an orientation

③ Do BFS (or DFS) to propagate orientation

④ Check every edge

③ and ④ can be done together



check this ed... fails!

neighbor cell
↳ give it same orientation.

Simplicial Complex

① Abstract Simplicial Cplx

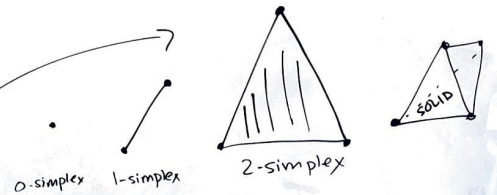
- Simplices: finite sets
- ASC = a finite set of simplices closed under taking subsets
- ex: $K = \{[a,b,c], [a,b], [b,c], [a,b],$ $[a], [b], [c], \emptyset\}$
  - needs all pairs & singletons.
  - some sources don't include this.

② Geometric S

- Simplices:
  of an af
  set of
- GSC =
  simplices
  ① If
  the
  o
  ② If

② Geometric Simplicial Cplx

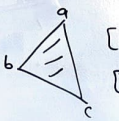
- Simplices: the convex hull of an affinely independent set of points in $\mathbb{R}^n$
- GSC = a finite set of simplices K such that
  1. If σ, τ are both in K, then either $\sigma \cap \tau = \emptyset$ or $\sigma \cap \tau$ is a face of both.
  2. If $\tau \leq \sigma$ and $\sigma \in K$, ("τ is a face of σ") then $\tau \in K$.



k simplex: defined by k+1 points

face: a simplex defined by a subset of vertices

coface: opposite of face



[abc] is a coface of [ab]
[ab] is a face of [abc]
[a] is a face of [ab]

facet: a face of codimension 1.

Q How/when can I use a simp cplx (abstract or geom.)? examples of how to use in data analysis

Q Relationship btwn ASC + GSC.

Simplicial Complex

① Abstract Simplicial Cplx

- Simplices: finite sets
- ASC = a finite set of simplices closed under taking subsets
- ex: $K = \{[a,b,c],[a,b],[b,c],[a,b],[a],[b],[c],\emptyset\}$
  - needs all pairs & singletons.
  - some sources don't include this.

easy: n vertices → pt to Std basis $\{e_i\}$ in $\mathbb{R}^m$

DW: "cannonical geometric realization"

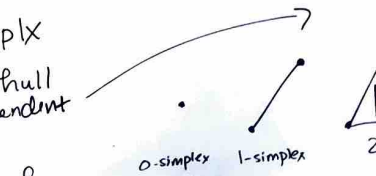
hard: Thm: an n-complex can be realized in $\mathbb{R}^{2n+1}$

note: $|K|$ is the underlying space (of a geometric realization of K)
ie- for an edge, every point is there.

FORGET WHERE VERTS ARE

② Geometric Simplicial Cplx

- Simplices: the convex hull of an affinely independent set of points in $\mathbb{R}^n$
- GSC = a finite set of simplices K such that
  ① If σ, τ are both in K, then either $\sigma \cap \tau = \emptyset$ or $\sigma \cap \tau$ is a face of both.
  ② If $\tau \leq \sigma$ ("τ is a face of σ") and $\sigma \in K$, then $\tau \in K$.



K simple

face: a si
by a

coface: opp

6

facet: a face

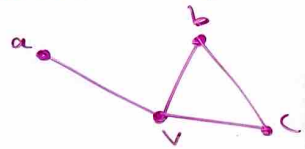
K = simp. cplx

$\sigma \in K$

Star $St(\sigma) := \{\sigma' \in K \mid \sigma \leq \sigma'\}$

Q: is this a simplicial cplx?

ex:



$St(v) = \{v, av, bv, cv\}$

No! $a, b, c \notin St(v)$.

Closed Star $\overline{St}(\sigma)$ = the closure of $St(\sigma)$

= add everything necessary to make $St(\sigma)$ a simp. cplx.

LINK $Lk(\sigma) = \overline{St}(\sigma) - St(\sigma)$

↑ in n-Manifolds,

$\overline{St} \cong \mathbb{B}^n$

$Lk \cong S^{n-1}$